

Pallet AMR Mechanisms & Components Report

This engineering document details the core mechanical systems, kinematics, operational logic, and specific component selections used in different versions of the low-profile Autonomous Pallet-handling Robot (APR) family (specifically E10T, E10 standard, and FL10).

For each mechanism, we outline **What it is**, **Why it is used**, **How it works**, and compile a **Component Specification Table**.

1. Telescopic Extending Fork Mechanism (Model: E10T)

What It Is

A multi-stage, motorized horizontal fork extension system that allows the robot's load-bearing forks to slide out forward from the lift carriage into the pallet openings.

Why It Is Used

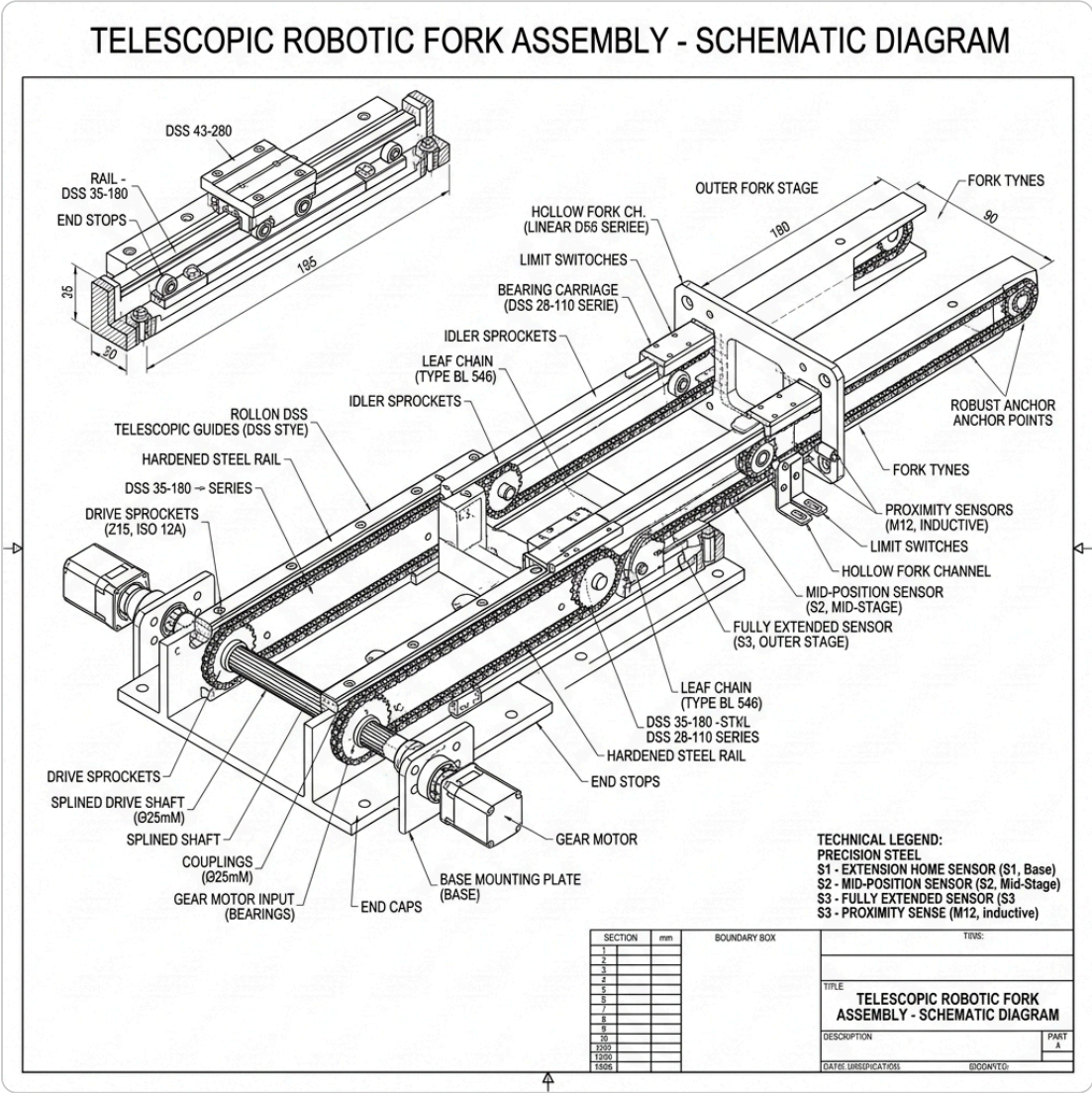
Traditional AGVs must drive their entire chassis forward to insert forks under a pallet, requiring large clearance areas. The telescopic extending fork allows the robot to remain stationary in a narrow aisle and slide only its forks forward under the pallet. This reduces the aisle width requirement for picking and placing to just `2.0 \text{ m}`.

How It Works

1. A single 24V 200W geared servo motor is mounted on the lift carriage.
2. This motor drives a transverse splined shaft.
3. Two chain drive sprockets slide along the splined shaft when the fork width adjusts, but rotate with it.

4. Each fork channel contains a dual-stage **Rollon DSS43** telescopic guide slide rail and a closed-loop leaf chain.
5. When the motor rotates the splined shaft, the chain loops pull the intermediate and outer stages of the telescopic slides forward, extending the forks by up to **1400\text{ mm}** (over-extension).

Diagram



Telescopic Robotic Fork Assembly Schematic Diagram

Components Used for Telescopic Fork Mechanism

Component Name	Manufacturer / Model	Specifications	Function / Purpose
Telescopic Slides	Rollon / DSS43-1170	Heavy-duty multi-stage linear slides; over-extension up to 1400mm; Q235 steel	Supports high vertical cantilever loads during fork extension
Geared Drive Motor	Leadshine / 24V 200W	Brushless DC geared servo motor; torque 2.5 Nm; integrated encoder	Actuates the transverse splined shaft to slide forks in/out
Splined Drive Shaft	Custom / Dia 25mm	High-tensile steel (40Cr), hardened spline profile	Transmits motor torque to sliding sprockets across adjustable width
Leaf Chains & Sprockets	local supplier / Type BL546	High-strength leaf chains, Z15 sprockets	Pulls the telescopic slide stages forward and backward
Proximity Sensors	Sick / M12 Inductive	4mm sensing range; NPN NO; IP67	Detects home, mid-stage, and full-extension positions

2. Retractable / Folding Load Roller Mechanism (E10T vs. E10 Standard)

What It Is

A pivot swingarm assembly at the front tip of each fork that retracts the load rollers vertically up into the fork body or deploys them down to the warehouse floor.

Why It Is Used

Double-sided pallets (such as EUR-2 and "田" shaped pallets) have closed bottom boards. A standard pallet wheel cannot roll over them without crashing.

- **During Fork Travel/Extension (Wheels UP):** The rollers remain folded **UP** inside the fork thickness (`65 \text{ mm}`) so the forks can slide over the bottom boards of the pallet without collision. The forks extend suspended in the air (cantilevered), carrying only their own light weight, which the telescopic slides support easily.
- **During Lifting & Transit (Wheels DOWN):** Once the forks are fully inserted (so the wheels are directly over a gap in the pallet's bottom boards), the wheels deploy **DOWN** to contact the floor. The lift carriage then raises, and the wheels support the heavy 1-ton payload, transforming the cantilevered forks into simple supported beams.

How It Works (Telescopic Forks vs. Fixed Forks)

Because the E10T forks extend telescopically by `1400 \text{ mm}` , a rigid mechanical pull-rod connecting the tip wheel to the chassis carriage is impossible (it would bind or restrict extension). Therefore, two different actuation methods are used depending on the AMR model:

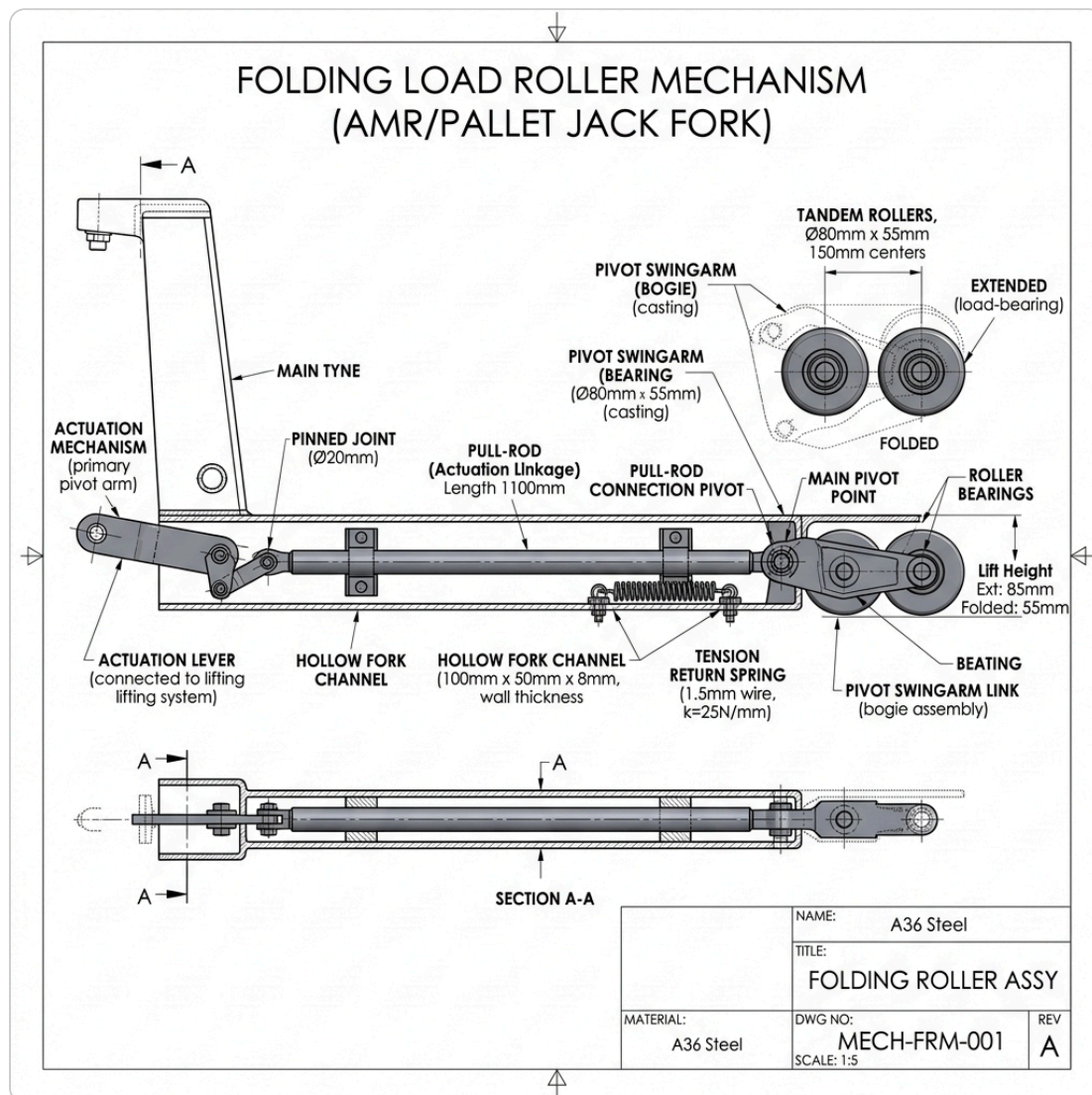
A. Telescopic Extending Fork (Model: E10T) - Motorized Actuation

1. A **compact, high-force 24V electric linear actuator** (or a miniature hydraulic cylinder) is mounted directly inside the tip of the outer fork stage.
2. Electrical power and control cables are routed from the main carriage to the actuator through a **flexible plastic drag chain (cable track)** that bends and nests inside the telescopic slide stages.
3. During extension, the actuator is kept fully retracted, holding the Bogie swingarm **UP** inside the fork cavity.
4. When the fork reaches full extension (detected by proximity sensors), the controller sends a 24V signal to the linear actuator, extending its piston shaft.
5. This pushes the bogie swingarm **DOWN** through the gaps in the bottom boards of the pallet until the polyurethane rollers contact the floor.

B. Fixed-Length Fork (Model: E10 Standard) - Passive Pull-Rod Linkage

1. Inside the hollow channel of each fixed fork runs a rigid **mechanical pull-rod**.
2. The front end of the pull-rod is pinned to the Bogie swingarm holding the tandem load rollers.
3. The rear end of the pull-rod is pinned to a lever on the vertical lift carriage frame.
4. When the carriage is at its lowest position, the linkage pulls the rod back, rotating the swingarm **UP** to fold the rollers inside the fork cavity.
5. As the carriage begins to rise, the linkage pushes the rod forward, forcing the swingarm to rotate **DOWN** to contact the floor.

Diagram



Folding Load Roller Mechanism Schematic Diagram

Components Used for Folding Wheel Mechanism

Component Name	Manufacturer / Model	Specifications	Function / Purpose
Tandem Load Rollers	Blickle / Dia 60mm	Polyurethane tread on steel core, needle bearings, 350kg capacity each	Support wheels that roll on the warehouse floor under the pallet
Pivot Swingarm (Bogie)	Custom Weldment	Cast steel (45#), precision-machined pivot holes	Houses the tandem rollers and rotates relative to the fork tyne
Electric Linear Actuator	local supplier / 24V DC	Compact high-force linear actuator; 50mm stroke; thrust 2500 N (E10T specific)	Mounts inside telescopic fork tip to push/pull swingarm directly
Flexible Drag Chain	Igus / E2 Micro series	Miniature plastic energy chain; minimum bending radius 28mm (E10T specific)	Routes actuator power and sensor cables inside telescopic stages
Actuation Pull-Rods	Custom Turnbuckle	High-tensile steel rods (M16 threads) for length calibration (E10 standard specific)	Transmits vertical carriage motion to the front pivot swingarms
Return Tension Springs	local supplier / 1.5mm wire	Spring steel, rate 25\text{N/mm} , length 150mm	Pulls the swingarms back into the folded position when unloaded
Pivot Pins & Bushings	Misumi / Hardened Steel	Dia 20mm, self-lubricating bronze bushings	High-load joints for swingarm and pull-rod/actuator linkages

3. Central Vertical Lift Carriage Mechanism (Models: E10, E10T)

What It Is

A single-motor vertical slide carriage mounted on the front face of the AMR chassis that raises and lowers the entire fork carrier assembly.

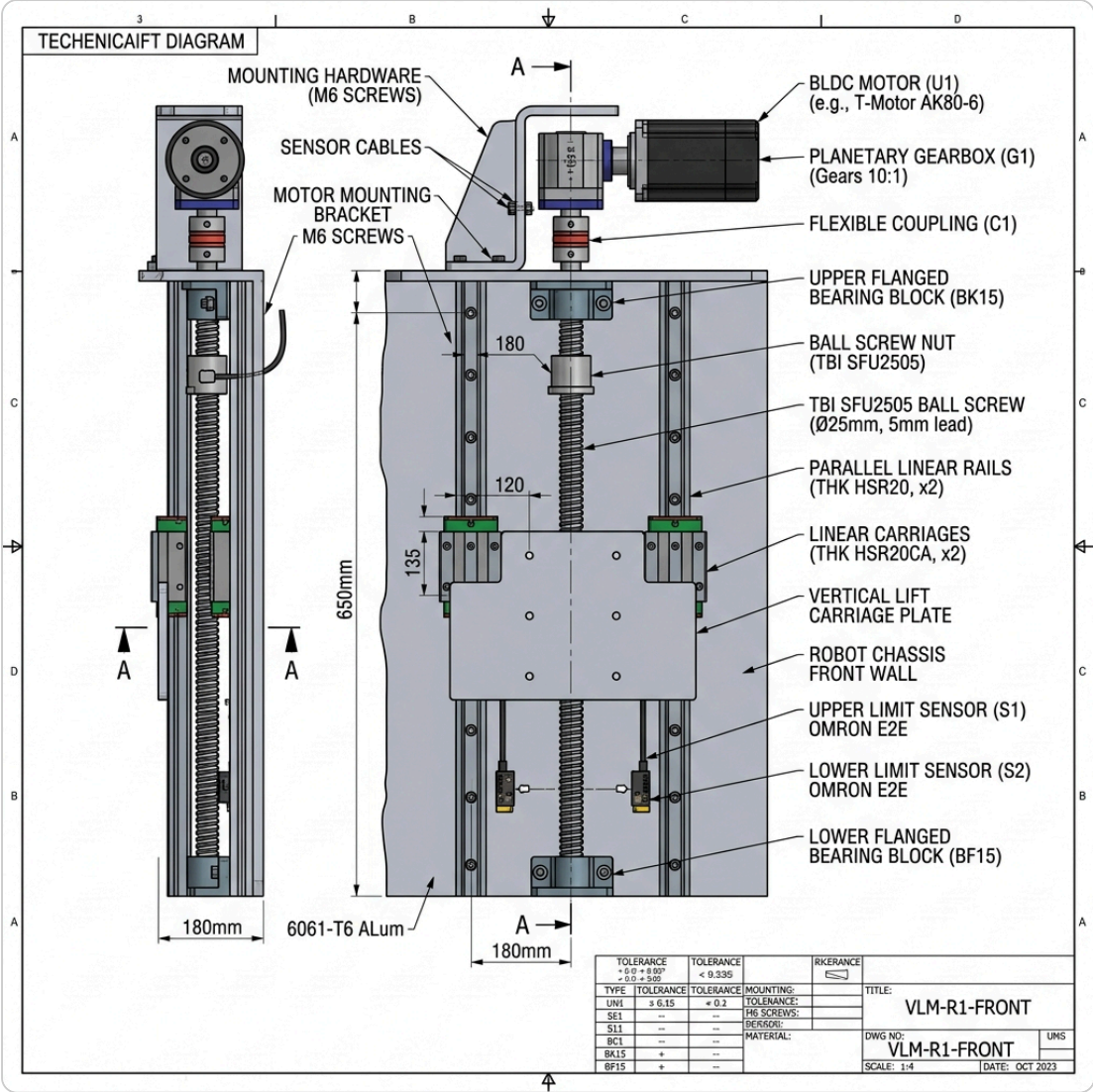
Why It Is Used

Rather than having separate scissor lifts and motors inside each fork (which increases weight, complexity, and fork thickness), the E10T uses a single central carriage lift. This keeps the forks extremely thin, eliminates the need for motor synchronization algorithms, and reduces cost.

How It Works

1. Two vertical linear profile rails are mounted on the front plate of the robot chassis.
2. The vertical lift carriage plate slides vertically along these rails.
3. A single vertical ball screw (TBI SFU2505, 5 \text{ mm} lead) is mounted centrally.
4. A 1000W BLDC motor with a 1:15 gearbox is mounted at the top, coupled to the ball screw.
5. When the motor rotates the ball screw, the ball nut pushes the carriage vertically, lifting both forks, the telescopic slides, and the load simultaneously.
6. High and low inductive proximity sensors detect limits.

Diagram



Vertical Lift Carriage Assembly Schematic Diagram

Components Used for Central Lift Mechanism

Component Name	Manufacturer / Model	Specifications	Function / Purpose
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Vertical
Linear
Guides

Hiwin / HGR25

Profile linear guide rails,
L=500mm, carbon steel

Provides rigid vertical
guidance and prevents
carriage twisting

Component Name	Manufacturer / Model	Specifications	Function / Purpose
Vertical Ball Screw	TBI Motion / SFU2505	Dia 25mm, 5mm lead, C7 accuracy, ball screw nut	Transforms motor rotation into high-force vertical linear motion
BLDC Lift Motor	Leadshine / 48V 1000W	Brushless DC motor; rated speed 3000 RPM; electromagnetic brake	Power source for lifting the 1-ton carriage assembly
Planetary Gearbox	local supplier / 1:15 ratio	Inline planetary gearbox; backlash < 8 arc-min; efficiency 95%	Multiplies motor torque to overcome vertical gravitational load
Carriage Plate	Custom CNC	8mm A36 steel plate, precision welded	Mounts the forks, width slides, and telescopic extenders
Limit Sensors	Omron / E2E Proximity	Cylindrical inductive sensors, NC/NO outputs	Detects vertical travel limits to prevent mechanical crash

4. Mast-Type Stacker Lift Mechanism (Model: FL10 Stacker)

What It Is

A nested double-stage vertical mast lift system similar to traditional forklift masts, used on the FL10 high-reach stacker.

Why It Is Used

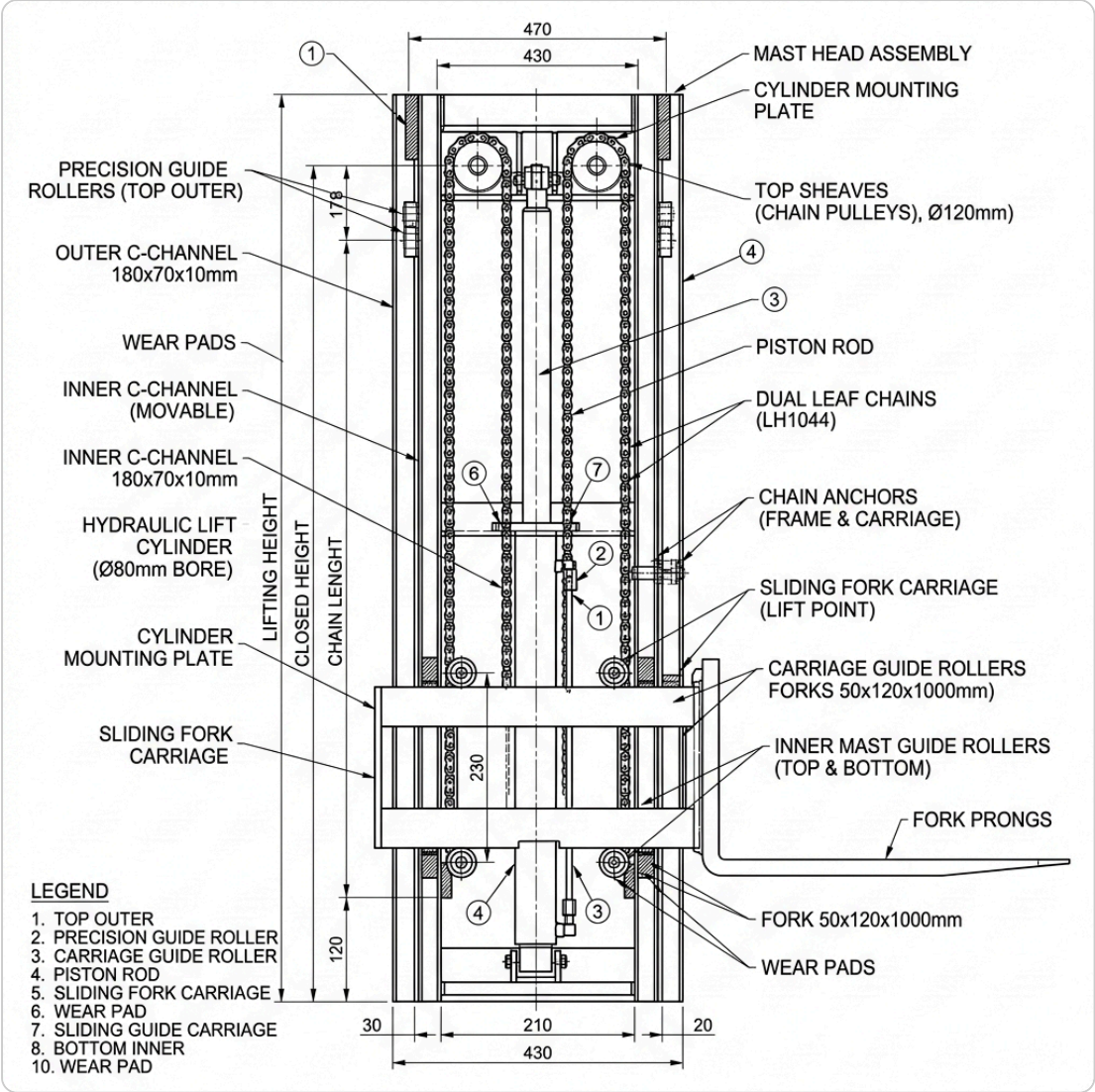
The central lift carriage (Section 3) is limited to a lift height of $\sim 100\text{ to }200\text{ mm}$ (just enough to clear the floor during transit). The FL10 requires lifting pallets up to

2.5 \text{ m} for double-layer stacking in warehouse racks. This requires a nested telescoping mast structure.

How It Works

1. Outer C-channel steel masts are fixed to the robot chassis.
2. Inner C-channel masts slide vertically inside the outer masts on sealed guide rollers.
3. A central hydraulic cylinder or an electric motor-driven leaf chain lift (LH1044) is actuated.
4. The piston rod pushes the inner mast upward.
5. Leaf chains anchored to the outer mast run over sheaves at the top of the inner mast and attach to the fork carriage.
6. This 2:1 mechanical ratio lifts the fork carriage at twice the speed of the inner mast extension, reaching a lift height of 2500 \text{ mm} .

Diagram



Nesting Mast Forklift Stacker Lifting Mechanism Schematic Diagram

Components Used for Mast Lift Mechanism

Component Name	Manufacturer / Model	Specifications	Function / Purpose
Nested Mast Channels	Custom / C-profile	High-strength steel Q345; 180\times70\times10\text{ mm} outer channel	Main structural guide channels for high-reach lifting

Component Name	Manufacturer / Model	Specifications	Function / Purpose
Leaf Chains	local supplier / LH1044	Heavy-duty forklift leaf chains; pitch 15.875mm; tensile strength 80 kN	Suspends the carriage and transmits the 2:1 lifting force
Lifting Cylinder	Custom / Hydraulic	80mm bore, 1200mm stroke, chrome-plated piston rod	Pushes the inner mast vertically (for hydraulic lift versions)
Mast Guide Rollers	local supplier / Sealed Ball	Dia 110mm, hardened outer race, double-sealed	Guide wheels that slide inside C-channels under bending moments
Chain Sheaves (Pulleys)	Custom / Dia 120mm	Cast iron core, deep groove, needle roller bearings	Guides the leaf chains at the top of the inner mast stage
Wear Pads	local supplier / Nylon-66	Self-lubricating wear plates, low friction	Prevents metal-to-metal friction between nested mast channels